d:

$$Y_{40} \longleftrightarrow \frac{1}{168}\sqrt{\frac{1}{\pi}}\left(35L_z^4 - 30L_z^2L^2 + 25L_z^2 - 6L^2 + 3(L^2)^2\right)$$

$$(Y_{44} + Y_{4-4}) \longleftrightarrow \frac{1}{168}\sqrt{\frac{35}{2\pi}}\left(L_+^4 + L_-^4\right)$$

$$(Y_{43} + Y_{4-3}) \longleftrightarrow -\frac{1}{24}\sqrt{\frac{5}{7\pi}}\left(L_z\left(L_+^3 + L_-^3\right) + \left(L_+^3 + L_-^3\right)L_z\right)$$

$$(Y_{42} + Y_{4-2}) \longleftrightarrow \frac{1}{168}\sqrt{\frac{5}{2\pi}}\left(\left(L_+^2 + L_-^2\right)\left(7L_z^2 - L^2 - 5\right) + \left(7L_z^2 - L^2 - 5\right)\left(L_+^2 + L_-^2\right)\right)$$

$$(Y_{41} + Y_{4-1}) \longleftrightarrow -\frac{1}{168}\sqrt{\frac{5}{\pi}}\left(\left(L_+ + L_-\right)\left(7L_z^3 - (3L^2+1)L_z\right) + \left(7L_z^3 - (3L^2+1)L_z\right)\left(L_+ + L_-\right)\right)$$

операторы Racah $\times \frac{1}{21}$

$$Y_{20} \longleftrightarrow -\frac{1}{42}\sqrt{\frac{5}{\pi}}\left(3L_z^2 - L^2\right)$$

$$Y_{21} + Y_{2-1} \longleftrightarrow \frac{1}{14}\sqrt{\frac{5}{6\pi}}\left(L_z\left(L_+ + L_-\right) + \left(L_+ + L_-\right)L_z\right)$$

$$Y_{22} + Y_{2-2} \longleftrightarrow -\frac{1}{14}\sqrt{\frac{5}{6\pi}}\left(L_+^2 + L_-^2\right)$$

операторы Racah $\times \frac{\sqrt{5}}{21}$

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p:

$$Y_{20} \longleftrightarrow -\frac{1}{2}\frac{1}{\sqrt{5\pi}}\left(3L_z^2 - L^2\right)$$

$$Y_{21} + Y_{2-1} \longleftrightarrow \frac{1}{2}\sqrt{\frac{3}{10\pi}}\left(L_z\left(L_+ + L_-\right) + \left(L_+ + L_-\right)L_z\right)$$

$$Y_{22} + Y_{2-2} \longleftrightarrow -\frac{1}{2}\sqrt{\frac{3}{10\pi}}\cdot\left(L_+^2 + L_-^2\right)$$

операторы Racah $\times \frac{1}{\sqrt{5}}$

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f:

$$Y_{60} \longleftrightarrow -\frac{1}{2376}\sqrt{\frac{1}{13\pi}}\left(231L_z^6 - (315L^2 - 735)L_z^4 + (105(L^2)^2 - 525L^2 + 294)L_z^2 - 5(L^2)^3 + 40(L^2)^2 - 60L^2\right)$$

$$(Y_{66} + Y_{6-6}) \longleftrightarrow -\frac{1}{144}\sqrt{\frac{7}{429\pi}}\left(L_+^6 + L_-^6\right)$$

...

Эквивалентные операторы (связаны с операторами Stevens и Racah)

Сферические гармоники из „Теория углового момента" Варшалович

$$Y_{00} = \frac{1}{\sqrt{4\pi}};$$

$$Y_{11} = -\frac{1}{2}\sqrt{\frac{3}{2\pi}}\sin\theta e^{i\varphi};\quad Y_{1-1} = \frac{1}{2}\sqrt{\frac{3}{2\pi}}\sin\theta e^{-i\varphi},\quad Y_{10} = \frac{1}{2}\sqrt{\frac{3}{\pi}}\cos\theta;$$

$$Y_{22} = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\sin^2\theta e^{2i\varphi};\quad Y_{2-2} = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\sin^2\theta e^{-2i\varphi};$$

$$Y_{21} = -\frac{1}{2}\sqrt{\frac{15}{2\pi}}\cos\theta\sin\theta e^{i\varphi};\quad Y_{2-1} = \frac{1}{2}\sqrt{\frac{15}{2\pi}}\cos\theta\sin\theta e^{-i\varphi};$$

$$Y_{20} = \frac{1}{4}\sqrt{\frac{5}{\pi}}\left(3\cos^2\theta - 1\right);$$

$$Y_{33} = -\frac{1}{8}\sqrt{\frac{35}{\pi}}\sin^3\theta e^{3i\varphi};\quad Y_{3-3} = \frac{1}{8}\sqrt{\frac{35}{\pi}}\sin^3\theta e^{-3i\varphi};$$

$$Y_{32} = \frac{1}{4}\sqrt{\frac{105}{2\pi}}\cos\theta\sin^2\theta e^{2i\varphi};\quad Y_{3-2} = \frac{1}{4}\sqrt{\frac{105}{2\pi}}\cos\theta\sin^2\theta e^{-2i\varphi};$$

$$Y_{31} = -\frac{1}{8}\sqrt{\frac{21}{\pi}}\left(5\cos^2\theta - 1\right)\sin\theta e^{i\varphi};\quad Y_{3-1} = \frac{1}{8}\sqrt{\frac{21}{\pi}}\left(5\cos^2\theta - 1\right)\sin\theta e^{-i\varphi};$$

$$Y_{30} = \frac{1}{4}\sqrt{\frac{7}{\pi}}\left(5\cos^2\theta - 3\right)\cos\theta;$$

$$Y_{44} = \frac{3}{16}\sqrt{\frac{35}{2\pi}}\sin^4\theta e^{4i\varphi};\quad Y_{4-4} = \frac{3}{16}\sqrt{\frac{35}{2\pi}}\sin^4\theta e^{-4i\varphi};$$

$$Y_{43} = -\frac{3}{8}\sqrt{\frac{35}{\pi}}\sin^3\theta\cos\theta e^{3i\varphi};\quad Y_{4-3} = \frac{3}{8}\sqrt{\frac{35}{\pi}}\sin^3\theta\cos\theta e^{-3i\varphi};$$

$$Y_{42} = \frac{3}{8}\sqrt{\frac{5}{2\pi}}\sin^2\theta\left(7\cos^2\theta - 1\right)e^{2i\varphi};\quad Y_{4-2} = \frac{3}{8}\sqrt{\frac{5}{2\pi}}\sin^2\theta\left(7\cos^2\theta - 1\right)e^{-2i\varphi}$$

$$Y_{41} = -\frac{3}{8}\sqrt{\frac{5}{\pi}}\sin\theta\left(7\cos^3\theta - 3\cos\theta\right)e^{i\varphi};\quad Y_{4-1} = \frac{3}{8}\sqrt{\frac{5}{\pi}}\sin\theta\left(7\cos^3\theta - 3\cos\theta\right)e^{-i\varphi}$$

$$Y_{40} = \frac{3}{16\sqrt{\pi}}\left(35\cos^4\theta - 30\cos^2\theta + 3\right)$$

Связь сферических гармоник и декартовых координат (Tesseral Harmonics)

$$Z_{00} = \frac{1}{\sqrt{4\pi}};$$

$$Z_{11}^{s} = \sqrt{\frac{3}{4\pi}}\left(\frac{y}{r}\right); \quad Z_{11}^{c} = \sqrt{\frac{3}{4\pi}}\left(\frac{x}{r}\right); \quad Z_{10} = \sqrt{\frac{3}{4\pi}}\left(\frac{z}{r}\right);$$

$$Z_{22}^{s} = \frac{1}{4}\sqrt{\frac{15}{\pi}}\left(\frac{2xy}{r^2}\right); \quad Z_{22}^{c} = \frac{1}{4}\sqrt{\frac{15}{\pi}}\left(\frac{x^2-y^2}{r^2}\right);$$

$$Z_{21}^{s} = \frac{1}{2}\sqrt{\frac{15}{\pi}}\left(\frac{yz}{r^2}\right); \quad Z_{21}^{c} = \frac{1}{2}\sqrt{\frac{15}{\pi}}\left(\frac{xz}{r^2}\right);$$

$$Z_{20} = \frac{1}{4}\sqrt{\frac{5}{\pi}}\left(3z^2 - r^2\right)/r^2;$$

$$Z_{33}^{s} = \sqrt{\frac{35}{32\pi}}\left(3x^2y - y^3\right)/r^3; \quad Z_{33}^{c} = \sqrt{\frac{35}{32\pi}}\left(x^3 - 3xy^2\right)/r^3;$$

$$Z_{32}^{s} = \sqrt{\frac{105}{16\pi}}\left(2xyz/r^3\right); \quad Z_{32}^{c} = \sqrt{\frac{105}{16\pi}}\left(x^2-y^2\right)z/r^3;$$

$$Z_{31}^{s} = \sqrt{\frac{21}{32\pi}}\left(y(5z^2-r^2)/r^3\right); \quad Z_{31}^{c} = \sqrt{\frac{21}{32\pi}}\left(x(5z^2-r^2)/r^3\right);$$

$$Z_{30} = \sqrt{\frac{7}{16\pi}}\left(z(5z^2-3r^2)/r^3\right);$$

$$Z_{44}^{s} = \frac{3}{16}\sqrt{\frac{35}{\pi}}\left(4(x^3y - xy^3)/r^4\right); \quad Z_{44}^{c} = \frac{3}{16}\sqrt{\frac{35}{\pi}}\left(x^4 - 6x^2y^2 + y^4\right)/r^4;$$

$$Z_{43}^{s} = \frac{3}{8}\sqrt{\frac{70}{\pi}}\left(3x^2y - y^3\right)z/r^4; \quad Z_{43}^{c} = \frac{3}{8}\sqrt{\frac{70}{\pi}}\left(x^3 - 3xy^2\right)z/r^4;$$

$$Z_{42}^{s} = \frac{3}{8}\sqrt{\frac{5}{\pi}}\left(2xy(7z^2-r^2)/r^4\right); \quad Z_{42}^{c} = \frac{3}{8}\sqrt{\frac{5}{\pi}}\left(x^2-y^2\right)(7z^2-r^2)/r^4;$$

$$Z_{41}^{s} = \frac{3}{4}\sqrt{\frac{5}{2\pi}}\left(yz(7z^2-3r^2)/r^4\right); \quad Z_{41}^{c} = \frac{3}{4}\sqrt{\frac{5}{2\pi}}\left(xz(7z^2-3r^2)/r^4\right);$$

$$Z_{40} = \frac{3}{16}\sqrt{\frac{1}{\pi}}\left(35z^4 - 30z^2r^2 + 3r^4\right)/r^4;$$

„s": $Y_{em} - Y_{e-m}$; „c": $Y_{em} + Y_{e-m}$